Learning useful tensor-network clusters

Meeting summary 23 October 2026

Goal and setup. Learn which original tensor cores to combine and which variables to leave open, so domain-conditioned contraction and GAC expose deductions and reduce DFS. Summaries are temporary; pruning is retained reversibly. The structural heuristic is generic, with no Sudoku-specific rules.

Current finding: useful learned clusters exist, but we have **not demonstrated a held-out 9×9 Sudoku learning advantage**. Some improvements over initialization occur on smaller problems. DFS counts below are totals within each stated dataset; lower is better.

1. **Supervised candidate scoring.** Predicted pruning, contraction time and contradiction for a fixed candidate list. Training loss fell **0.607 → 0.043** over 200 epochs. On the default 100 Sudoku puzzles: **237 nodes**, versus plain MAC 245 and heuristic 177. This benchmark overlaps training; it does not establish generalization or learn construction.
2. **Early constructive REINFORCE.** Pruning reward with time cost and annealed structural guidance failed to learn useful construction. Fixed guidance with a treewidth cost produced useful samples, but used **232 nodes** on the same overlapping benchmark. Repeated application still used 232. Weaker guidance alone also failed to improve its untrained control.
3. **Teacher imitation + mixed-reward RL.** Added partial-cluster context, target-first construction, measured teachers and pruning/search rewards. Development best: **295 nodes**, versus plain 323 and heuristic 194. On **10 fresh held-out Sudoku puzzles**: trained **455**, random initialization 449, plain 475, heuristic 245. No root solved without DFS. The development gain did not establish a held-out learning gain; imitation and RL effects were not isolated.
4. **Easier 4×4 curriculum.** Ordinary GAC already solved all 588 representative minimal unique clue puzzles. We therefore used partial-domain states with remaining pruning opportunities. On **32 reserved states**: plain 123 nodes, ordinary training 119, repair imitation 122, value-aware encoder + repair 119, heuristic 115. Value-aware training closed **24.4%** of the removable primary-domain gap, versus heuristic 49.0%. These states have multiple solutions.
5. **Better teachers and prefix training.** Generic failed-value probes supplied verified contraction witnesses, including 80 useful Sudoku training examples; 41 used 33–64 cores at less than 2,000 estimated work. Teaching useful completions from the learner's own prefixes improved a tiny generic probe from **5/8 to 8/8** greedy successes. Broader topology transfer remained weak; nonrelational coupled-graph training did not beat its initialization.
6. **Relational representation.** Added value/tuple context to preserve correlations lost by pooling. On **48 fresh generic graphs**: initialization 95 nodes, teacher imitation 93, completion imitation **91**, uniform construction 87, heuristic 48. This shows a small learned gain, below simple controls. Zero-shot transfer to already-used Sudoku puzzles gave no root pruning.
7. **Equal attempts and wider inference.** With frozen weights and no cumulative work cap, raising both methods' allowances from 32 to 128 to 512 reduced learned generic-graph nodes **71 → 50 → 48**, matching heuristic 48. Sudoku4 stayed **119**; heuristic improved 115 → 114. These reused examined datasets, with no retraining: more attempts helped the graph family, not that Sudoku policy.
8. **DFS-savings-only REINFORCE.** Rewarded actual completed-search node savings, removing pruning/time bonuses. Eight epochs did not improve the selected checkpoint: **initialization remained best**. As forced teacher prefixes were phased out, node-saving trials fell from 195 to zero in epoch 7, then four in epoch 8. Validation timed out, so aggregate DFS totals were unavailable.
9. **Maze/Dai-style Q-learning with replay.** Sequential construction, terminal pruning reward, undiscounted backups and successful/failed replay; no forced prefixes. Tiny generic smoke: greedy pruning learned and nodes fell **2 → 1** with one attempt, tying the heuristic. Latest saved Sudoku run: **6/8 planned epochs**; Q loss 0.0171 → 0.0118, but greedy usefulness only 0–1/17. Full validation completed 20–21/25 solves; no comparable DFS total, and initialization remained selected.

Evidence: saved supervised and RL logs/checkpoints; benchmark `results/raw/mac_sudoku*.out`; RL reports in `prior_probe_20261021`, `useful_clusters_20261021`, `efficiency_curriculum_20261021` and `node_budget_sweep_20261022`. Latest-run figures are a saved-checkpoint snapshot, not a completed new held-out test.